

Group Project

CRISPR Screening and Analysis

Online learning week

28 July - 1 August

 wellcome connecting science		CRISPR Screening and Analysis - Genome Informatics and Computational Design and Analysis 28 July - 1 August 2025, virtual via Zoom				 wellcome sanger institute	UNIVERSITY OF WESTMINSTER	
PRECOURSE ON LMS - Linux and R (for interest only)	Time	Monday 28 July	Tuesday 29 July	Wednesday 30 July	Thursday 31 July	Friday 1 August	Time	
	08:45	Join Call	Join Call	Join Call	Join Call	Join Call	08:45	
	09:00	Course introduction session Wellcome Connecting Science, training team, participants	Daily recap	Daily recap	Daily recap	Group project reminder Matthew Coelho and Kalpana Surendranath	09:00	
	09:15		Genome engineering workflows and screening Kalpana Surendranath	Applications of base editing and prime editing in pooled and arrayed screens Matthew Coelho	ENCORE Eloise Lockyer		09:15	
	09:30					Group work time	09:30	
	09:45						09:45	
	10:00						10:00	
	10:15						Break	Break
	10:30	10:30						
	10:45	10:45						
	11:00	Introduction to the CRISPR toolbox Andrew Bassett	Case studies Munuse Savash Kalpana Surendranath	Base editing guide RNA design Cansu Dincer	CRISPR screening informatics Jamie Billington Saamin Cheema	Group work time		
	11:15						11:15	
	11:30	11:30						
	11:45	11:45						
	12:00	Guided computational practical: genes and genomes Saroor Patel	Q&A	Q&A	Q&A		12:00	
	12:15						12:15	
	12:30						12:30	
	12:45		Lunch	Lunch	Lunch		Lunch	12:45
	13:00	13:00						
	13:15	13:15						
	13:30	13:30						
	13:45	Refresher quiz Saroor Patel	CRISPResso and validation of gene-edited models Khalid Akram	SNV and knock-in validation (Sanger sequencing - TIDE) Sandeep Rajan	Seminar Manuel Kaulich	Group presentations: Groups of 6 10 min presentation/ group and 5min QA	13:45	
	14:00						14:00	
	14:15	Seminar Alice Refermat			Refresher quiz Jamie and Saamin		14:15	
	14:30						14:30	
	14:45						14:45	
	15:00		15:00					
	15:15	Intro to group project Matthew Coelho, Kalpana Surendranath	Refresher quiz Kalpana and Khalid	Refresher quiz Cansu and Sandeep	Project and practical Q&A		Wrap up and close	15:15
	15:30							15:30
	15:45							15:45
		Monday 28 July	Tuesday 29 July	Wednesday 30 July	Thursday 31 July		Friday 1 August	

Virtual Groups

Group 1
Mehmet Bikkul
Ruhi Anjum
Zhe Yang
Marina Gysin
Katherine Diane Lugo Bardales

Group 2
Melanie Cook
Naima Azouzi
Claudia Maria Pinna
Kayani Kayani
Bo Fussing

Group 3
Andrew Curtis
Anam Razzak
Snir Boniel
Gratiela Gradisteanu
Salwa Lin
Ilaria Persico

Group 4
Bryony Kennedy
Zainab Bashir
Olayiwola Popoola
Marcos Magalhães
Mona Mahfood

Group 5
Iulia Maria Sabau
David Twesigomwe
Luigi Celauro
Shalini Kamu Reddy
Fatima Mohamedelmugadam

Group Project

A missense mutation in the coding region of the **SRY gene** (**g.2787144A>G, p.C154R**) has recently been found to alter the expression of the protein. This leads to an uncommon disorder in men.

The presence of the mutation does not always lead to disease, suggesting there are other factors at play. Your research team have been tasked with identifying genes involved in disease progression - particularly new drug targets to reverse the disorder.

Groups' tasks



- 1) Describe a CRISPR-based workflow to introduce the mutation in *SRY* in an appropriate cell model and verify the mutation.
- 2) Describe how you would design a CRISPR KO screen to identify modulators of *SRY* expression in this modified cell model.
- 3) Describe next steps after the screen, including a validation strategy.

*For each step, give a **rationale** for your approach and describe the **analysis** in each case.*



Modelling the *SRY* Promoter Mutation - Pointers

- What is your model system?
- Should it naturally express SRY? How about the efficiency of genetic manipulation?
- Your choice of CRISPR tools to introduce a precise genetic alteration
- Optimal method to provide correct copies of the DNA? How would you deliver the tools into the cells?
- What methods will allow you to screen for the correction?
- What methods can be used to measure both RNA and protein levels?

CRISPR Knockout Screen - Pointers



- How will you ensure the mutation is present? What factors will limit your analysis of more genes?
- What readout will allow you to separate cells with higher or lower expression?
- How will you sort and collect the most interesting/informative population of cells?
- What sequencing strategy will allow you to identify sgRNAs?
- How will you leverage the sequencing data to find candidate genes worth investigating further?

Group presentation

15 min per group

10 min for presentation and 5 min for questions

All the best!